Template-free optimal pulse processing for saturating microcalorimeters: shift-invariant whitened PCA manifolds and Woodbury-accelerated non-stationary maximum-likelihood estimation

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Abstract

The optimal filter used to estimate energies of microcalorimeter pulses assumes a fixed template that scales linearly with energy, stationary noise and a known arrival time. Transition-edge sensors (TES) and other thermal detectors violate all three: pulses change shape as the sensor saturates, the noise changes during the pulse, and the arrival time is random at the sub-sample level. We present a three-stage estimator that addresses these effects without a calibrated template. (A) A shift-invariant principal component analysis in the noise-whitened domain learns an orthonormal pulse subspace and sub-sample arrival times by classification EM. The E-step is a phase-coherent cross-correlation and the M-step is a singular value decomposition of counter-rotated records; both are exact because spectral whitening commutes with Fourier shifts. (B) The aligned coefficients are fitted by a smooth spline manifold through the origin, parameterised by whitened arc length, which equals the cumulative matched-filter signal-to-noise ratio and has unit Fisher information. (C) Pulse-correlated noise and residual shape fluctuations are modelled as a low-rank, energy-dependent covariance $R(E, t_0) = R_0 + T_{t_0}\, G\, \Sigma(E)\, G^{\mathsf{T}} T_{t_0}^{\mathsf{T}}$. The Woodbury identity and the matrix determinant lemma reduce the non-stationary log-likelihood to an expression in $K_b \approx 10$ dimensions, evaluated at the cost of a stationary filter with K_b templates instead of the $\mathcal{O}(N^3)$ of a dense non-stationary filter. We prove that shape fluctuations along the manifold tangent cannot be told apart from energy fluctuations in unlabelled continuum data; they can be identified from mono-energetic clusters, whose energies need not be known. On a physical Euler–Maruyama simulation of a saturating TES, with the model learned on one spectrum and tested on lines at different energies, the estimator reaches 1.7–2.8 eV FWHM from 1 to 11 keV when the absorber thermalisation time varies by ±2%, against 2.0–12.4 eV for the optimal filter. This is within 0.03 eV of a per-line oracle Gaussian model. Without that morphology variation, the optimal filter is already close to optimal and the gain shrinks to at most 15%: most of the large improvement comes from modelling shape fluctuations, not from the non-linearity itself. Without morphology variation, arrival times reach the Cramér–Rao bound once a deterministic sampling-phase bias is removed. That bias is the remaining timing limitation. We discuss which claims of calibration independence survive: the manifold and the noise model are learned without labels, but mapping the manifold coordinate to absolute energy still requires reference lines. An open-source Python implementation accompanies the paper.

1 Introduction

Thermal microcalorimeters such as transition-edge sensors (TES), metallic magnetic calorimeters and thermal kinetic-inductance detectors reach energy resolutions of a few eV at keV energies [1, 2]. The standard estimator is the optimal filter (OF) [3, 4]: a frequency-domain matched filter that is the minimum-variance unbiased amplitude estimator if (i) every pulse is a scaled copy of one template, (ii) the noise is additive, Gaussian and stationary, and (iii) the arrival time is known. Real detectors break each assumption. As a TES is driven up its transition, the pulse

shape evolves with energy and eventually saturates. The Johnson and thermal-fluctuation noise depend on the instantaneous resistance and temperature, so the noise during a pulse differs from the baseline noise. Arrival times are continuous, which leads to sampling-phase ("arrival-time") effects [4]. Position-dependent thermalisation adds shape variation that is not a function of energy.

Two lines of work have attacked these problems. Physics- or calibration-based non-linear filters fit energy-dependent templates and use energy-dependent, non-stationary covariances [5–8]. Fixsen *et al.* [7] gave this a differential-geometric interpretation: the noise-free pulses form a curve in record space and the optimal estimate is a projection onto that curve under the local noise metric. These methods rely on calibrated templates (from a physical model or a multi-photon calibration source), and the time-domain non-stationary covariance costs $\mathcal{O}(N^3)$ to factorise for every energy. Data-driven methods use principal component analysis (PCA) of the records [9–12]. In earlier work [12] we proposed a template-free estimator based on a weighted EM-PCA [13, 14] of a phase-differenced Fourier transform that is invariant to arrival time. That construction has two weaknesses. It discards arrival-time information by design. Its pulse-height estimator, the norm of the coefficient vector, is rectified: it is biased upward at low signal-to-noise ratio, like the amplitude of a Rician variable.

This paper presents a successor that keeps the template-free spirit of Ref. [12] and replaces each heuristic with a likelihood-based component. The contributions are:

1. **Shift-invariant whitened PCA** (section 3). We show that the symmetric spectral whitening operator $R_0^{-1/2}$ commutes with fractional Fourier shifts, whereas a Cholesky factor of the Toeplitz covariance does not (lemma 1). In the whitened domain, alignment (E-step) and subspace learning (M-step) are then exact operations on the data. We prove monotone convergence (proposition 2) and show that the noise does not bias the learned subspace (proposition 1). We also identify a degeneracy: a subspace that contains the time derivative of the pulse carries no timing information (proposition 3).

2. **Arc-length spline manifold** (section 4). The aligned coefficients are fitted by a cubic P-spline graph through the origin. In whitened arc length s, the Fisher information of s is one, and s is unbiased at zero energy, which removes the rectification of the coefficient norm (proposition 4).

3. **Woodbury-accelerated non-stationary ML** (section 5). With a low-rank, energy-dependent excess covariance, the exact non-stationary log-likelihood reduces to $K_b \ll N$ dimensions (theorem 1). We prove an identifiability result (proposition 5): fluctuations along the manifold tangent cannot be learned from unlabelled continuum events, but can be learned from mono-energetic clusters, without knowing their energies.

4. **A physical simulation study** (sections 6 and 7) based on an Euler–Maruyama integration of the electro-thermal Langevin equations of a saturating TES, with held-out test lines. We compare against the OF and the PCA norm, and against Cramér–Rao bounds and an oracle Gaussian model.

5. **Open-source software**: a tested Python package with a scikit-learn-like interface (appendix A).

2 Signal model and notation

A record $\boldsymbol{y} \in \mathbb{R}^N$ sampled at rate f_s is modelled as

$$\boldsymbol{y} = T_{t_0}\, \boldsymbol{m}(E, \eta) + \boldsymbol{n}, \tag{1}$$

where $\boldsymbol{m}(E, \eta)$ is the noise-free pulse for deposited energy E and nuisance morphology η (e.g. absorption position), t_0 is the arrival time in samples, and $\boldsymbol{n}$ is zero-mean Gaussian noise. Let F be the unitary DFT. The (band-limited, circular) shift operator is

$$T_\tau = F^H \operatorname{diag}\left(e^{-2\pi i k \tau / N}\right) F, \qquad (T_\tau \boldsymbol{x})(t) = x(t - \tau), \tag{2}$$

with the Nyquist bin removed so that T_τ maps real signals to real signals and is orthogonal on the remaining band for any real τ. Its generator is the spectral derivative $D = F^H \operatorname{diag}(2\pi i k/N) F$, with $T_\tau = e^{-\tau D}$ and $D^\mathsf{T} = -D$. Baseline noise is approximated by a circulant covariance $R_0 = F^H \Lambda F$, with $\Lambda = \operatorname{diag}(\lambda_k)$ estimated from pulse-free records as $\lambda_k = \mathbb{E}|Y_k|^2/N$. This approximation is asymptotically equivalent to the Toeplitz covariance [15] and is standard in OF practice [3]. We exclude the DC bin (baseline offset) and the Nyquist bin, which leaves $N - 2$ real degrees of freedom.

Lemma 1 (Whitening commutes with shifts)**.** *Let $W = R_0^{-1/2} = F^H \Lambda^{-1/2} F$. Then $W T_\tau = T_\tau W$ for all $\tau \in \mathbb{R}$, and $W\boldsymbol{n}$ is white with unit variance on the analysis band. If G is any other whitening matrix ($G R_0 G^\mathsf{T} = I$), then $G = QW$ with Q orthogonal, and G commutes with all T_τ if and only if Q is diagonalised by F. In particular, the inverse Cholesky factor L_0^{-1} of a non-circulant Toeplitz R_0 does not commute with shifts.*

Proof. W and T_τ are both diagonal in the Fourier basis. For the second statement, $(GW^{-1})(GW^{-1})^\mathsf{T} = G R_0 G^\mathsf{T} = I$, so $Q = GW^{-1}$ is orthogonal, and Q commutes with the full family $\{T_\tau\}$ iff it is a function of D, i.e. diagonal in F. A lower-triangular L_0^{-1} is not circulant unless R_0 is diagonal. □

All subsequent processing works with whitened records $\boldsymbol{y}_\mathrm{w} = W\boldsymbol{y}$. A delay of the raw record is the same delay of the whitened record, so a shifted pulse remains a shifted pulse after whitening. This is the property that makes the methods below exact rather than approximate.

3 Stage A: shift-invariant whitened PCA

3.1 Model and objective

In the whitened domain we model the pulses of a training set as lying in an unknown K-dimensional subspace with orthonormal basis $B \in \mathbb{R}^{K \times N}$ ($BB^\mathsf{T} = I$):

$$\boldsymbol{y}_\mathrm{w}^{(j)} = T_{\tau_j} B^\mathsf{T} \boldsymbol{a}_j + \boldsymbol{\varepsilon}_j, \qquad \boldsymbol{\varepsilon}_j \sim \mathcal{N}(0, I). \tag{3}$$

The amplitudes $\boldsymbol{a}_j$ profile out as $\hat{\boldsymbol{a}}_j = B T_{-\tau_j} \boldsymbol{y}_\mathrm{w}^{(j)}$. The negative log-likelihood is then $\sum_j \|\boldsymbol{y}_\mathrm{w}^{(j)}\|^2 - J(B, \{\tau_j\})$ with captured power

$$J(B, \{\tau_j\}) = \sum_j \left\| B\, T_{-\tau_j} \boldsymbol{y}_\mathrm{w}^{(j)} \right\|^2 . \tag{4}$$

We maximise J by alternating over the latent shifts and the basis. This is a hard (classification) EM, a shift-invariant analogue of EM-PCA [13, 16].

3.2 E-step: phase-coherent cross-correlation

For fixed B, each τ_j maximises $g_j(\tau) = \|\boldsymbol{x}_j(\tau)\|^2$, where $x_{jk}(\tau) = \langle \boldsymbol{b}_k, T_{-\tau} \boldsymbol{y}_\mathrm{w}^{(j)} \rangle$. Writing $\hat{\cdot}$ for the real FFT and $w_k \in \{1, 2\}$ for the multiplicity of each half-spectrum bin,

$$x_{jk}(\tau) = \frac{1}{N} \sum_{k'} w_{k'}\, \Re\left[\overline{\hat{b}_{k,k'}}\, \hat{y}_{j,k'}\, e^{2\pi i k' \tau / N} \right] . \tag{5}$$

For all integer lags this is a cross-correlation, computed with one inverse FFT per basis vector. The maximum is then refined to sub-sample precision with guarded Newton steps on the exact band-limited function (5). Its τ-derivatives follow analytically by multiplying the summand by $(2\pi i k'/N)^m$, at $\mathcal{O}(KN)$ cost per step. Unlike up-sampling or parabolic interpolation, this optimises the true likelihood, so the estimate inherits the asymptotic efficiency of maximum likelihood.

3.3 M-step: counter-rotation and SVD

For fixed $\{\tau_j\}$, J is maximised by the top-K right singular vectors of the matrix X whose rows are the counter-rotated records $T_{-\tau_j}\boldsymbol{y}_{\mathrm{w}}^{(j)}$. The counter-rotation is a phase ramp in the frequency domain. We compute the $N \times N$ scatter matrix $X^{\mathsf{T}}X$ and its eigendecomposition.

Proposition 1 (Whitened noise does not bias the subspace)**.** *If $\boldsymbol{y}_{\mathrm{w}}^{(j)} = T_{\tau_j}\boldsymbol{\mu}_j + \boldsymbol{\varepsilon}_j$ with $\boldsymbol{\varepsilon}_j$ white and the shifts are known, then $\mathbb{E}[X^{\mathsf{T}}X]/n = \frac{1}{n}\sum_j \boldsymbol{\mu}_j\boldsymbol{\mu}_j^{\mathsf{T}} + P$, where P projects onto the analysis band. The noise term is isotropic, so the eigenvectors are those of the noise-free scatter matrix.*

Proof. $T_{-\tau_j}$ is orthogonal and commutes with P, so $\mathbb{E}[T_{-\tau_j}\boldsymbol{\varepsilon}_j\boldsymbol{\varepsilon}_j^{\mathsf{T}}T_{-\tau_j}^{\mathsf{T}}] = P$. □

The proposition fails without whitening: coloured noise adds a non-isotropic term R_0 and tilts the subspace toward the noise-dominated frequencies. With *estimated* shifts, low-SNR records are aligned partly to their own noise. This is the "template from noise" bias known from cryo-electron microscopy [17, 18]. We therefore learn B only from records whose whitened excess power exceeds $\mathrm{SNR}_{\min}^2$. All records are still aligned and projected.

3.4 Convergence, gauge and a timing degeneracy

Proposition 2 (Monotonicity)**.** *Let the E-step maximise $J_r = \sum_j \|B_{1:r}T_{-\tau_j}\boldsymbol{y}_{\mathrm{w}}^{(j)}\|^2$ over τ_j for the leading $r \le K$ basis vectors, and let the M-step take the top-K singular vectors. Then J_r is non-decreasing over the iterations.*

Proof. The E-step maximises J_r for fixed B. The top-r right singular vectors maximise $\sum_j \|B_{1:r}\boldsymbol{x}_j\|^2$ over all r-frames (Ky Fan), and they are the first r of the top-K singular vectors. □

The model is invariant under a common shift $(B, \tau_j) \mapsto (T_\delta B, \tau_j + \delta)$. We fix this gauge by setting the median arrival time of the training set to zero after every iteration; this leaves J_r unchanged. The choice of r is governed by the following result.

Proposition 3 (A subspace containing the pulse derivative carries no timing information)**.** *Let P_B be the projector onto* span(B) *and $g(\tau) = \|P_B T_{-\tau}\boldsymbol{y}\|^2$. At a point where $\boldsymbol{z} = T_{-\tau}\boldsymbol{y} \in$* span$(B)$*,*

$$g''(\tau) = -2\,\|(I - P_B)\,D\boldsymbol{z}\|^2. \tag{6}$$

The curvature of the profile likelihood in τ, and so the Fisher information of the subspace estimator of t_0, comes only from the part of the time derivative of the pulse that lies outside *the subspace.*

Proof. $\frac{d}{d\tau}\boldsymbol{z} = D\boldsymbol{z}$. Then $g' = 2\boldsymbol{z}^{\mathsf{T}}P_B D\boldsymbol{z}$ and $g'' = 2(D\boldsymbol{z})^{\mathsf{T}}P_B(D\boldsymbol{z}) + 2\boldsymbol{z}^{\mathsf{T}}P_B D^2\boldsymbol{z}$. With $P_B\boldsymbol{z} = \boldsymbol{z}$ and $D^{\mathsf{T}} = -D$, the second term equals $-2\|D\boldsymbol{z}\|^2$, and $\|D\boldsymbol{z}\|^2 - \|P_B D\boldsymbol{z}\|^2 = \|(I-P_B)D\boldsymbol{z}\|^2$. □

A free K-dimensional subspace learned from jittered data tends to absorb the derivative direction, since it explains small shifts linearly, and its E-step then becomes ill-posed. We observed exactly this for $K \ge 3$ with $r = K$ (timing scatter of a full sample). With $r = 1$ the alignment uses only the leading, mean-pulse direction $\boldsymbol{b}_1$. Since $\langle \boldsymbol{b}_1, D\boldsymbol{b}_1\rangle = 0$, proposition 3 then gives the classical OF timing information $a_1^2\|D\boldsymbol{b}_1\|^2$. The remaining $K-1$ directions describe shape variation *after* alignment. The final timing estimate comes from the manifold-constrained likelihood of Stage C, where the pulse family is one-dimensional and cannot absorb shifts.

4 Stage B: arc-length spline manifold

4.1 Representation

The aligned coefficients $\boldsymbol{a}_j = BT_{-\tau_j}\boldsymbol{y}_{\mathrm{w}}^{(j)} \in \mathbb{R}^K$ have unit isotropic noise (proposition 1) and concentrate near a curve, the image of the energy axis. Let $\boldsymbol{e}$ be the leading uncentred principal direction of $\{\boldsymbol{a}_j\}$ (the mean-pulse direction), $u = \langle \boldsymbol{e}, \boldsymbol{a}\rangle$, and let the columns of $\tilde{W}$ span the orthogonal complement. We represent the curve as the graph

$$\boldsymbol{c}(u) = \boldsymbol{e}\,u + \tilde{W}\,u\,\boldsymbol{h}(u), \qquad \boldsymbol{h} = \text{cubic P-spline}, \tag{7}$$

fitted by penalised least squares [19]. The penalty is selected by generalised cross-validation [20]. Three properties are built in. The curve passes through the origin (a zero-energy event is no pulse) and is linear for small signals. It is continued linearly for $u < 0$ and tangentially beyond the data. Being a graph over u, it cannot fold back on itself. Free-form principal curves [21], which we tried first, did fold back in sparse, strongly saturated regions.

4.2 Arc length and the Fisher information

We re-parameterise by whitened arc length, $s(u) = \int_0^u \|\boldsymbol{c}'(v)\|\,dv$, so that $\|d\boldsymbol{c}/ds\| = 1$. Under stationary noise with known alignment, $\boldsymbol{a} = \boldsymbol{c}(s) + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, I_K)$, the Fisher information is $I_{ss} = \|d\boldsymbol{c}/ds\|^2 = 1$, so

$$\sigma_s \geq 1, \qquad \sigma_E \geq \left|\frac{dE}{ds}\right|, \qquad \mathrm{FWHM}_E \geq 2.355\left|\frac{dE}{ds}\right|. \tag{8}$$

In the linear regime, $\boldsymbol{c}(s) = s\boldsymbol{e}$ and $s = \langle \boldsymbol{e}, \boldsymbol{a}\rangle$ is the OF amplitude divided by its baseline resolution, i.e. the matched-filter SNR. The arc length is therefore the natural generalisation of the OF SNR to evolving pulse shapes. The ML estimate given the alignment is the closest point of the curve.

When the arrival time is estimated jointly, $\partial_s\boldsymbol{\mu} = B^{\mathsf{T}}\boldsymbol{c}'$ and $\partial_\tau\boldsymbol{\mu} = -D\boldsymbol{\mu}$ with $\boldsymbol{\mu} = B^{\mathsf{T}}\boldsymbol{c}(s)$. In the linear regime $\langle \boldsymbol{c}', D\boldsymbol{\mu}\rangle \propto \langle \boldsymbol{b}_1, D\boldsymbol{b}_1\rangle = 0$ and energy and time are Fisher-orthogonal. When the shape evolves with energy they are not: the cross term is the information-geometric origin of time walk, and the CRB on s inflates by $1/(1-\rho^2)$, where ρ is the Fisher correlation coefficient. The joint fit of Stage C accounts for this automatically.

4.3 No rectification at low energy

Proposition 4 (Rectification of the coefficient norm)**.** *If $\boldsymbol{a} \sim \mathcal{N}(\boldsymbol{c}, I_K)$ then $\mathbb{E}\|\boldsymbol{a}\|^2 = \|\boldsymbol{c}\|^2 + K$ and, for $\boldsymbol{c} = 0$, $\|\boldsymbol{a}\|$ follows a χ_K distribution with mean $\sqrt{2}\,\Gamma(\frac{K+1}{2})/\Gamma(\frac{K}{2}) > 0$. The manifold coordinate with linear continuation through the origin satisfies $\hat{s} = \langle \boldsymbol{e}, \boldsymbol{a}\rangle \sim \mathcal{N}(0,1)$ for $\boldsymbol{c} = 0$, to the extent that the curve is straight within a few noise units of the origin, and is unbiased.*

The norm estimator of Ref. [12] therefore has an energy-dependent bias of order $K/(2\|\boldsymbol{c}\|)$ that vanishes only at high SNR, plus a non-linear response near threshold. The manifold coordinate has neither.

4.4 Energy scale

The manifold, and hence s, is learned without labels. The map $E(s)$ is monotone and, in the linear regime, proportional, but its curvature at high energy cannot be recovered from pulse shapes alone. We fit it with a shape-preserving cubic interpolant through the origin and the positions of known lines, as for any non-linear detector. This is the sense in which the method is template-free but not calibration-free.

5 Stage C: Woodbury-accelerated non-stationary maximum likelihood

5.1 Covariance model

Pulse-correlated noise has several sources: resistance- and temperature-dependent Johnson and phonon noise, residual shape fluctuations such as position dependence, and the shape distortion caused by sampling a pulse that is not band-limited at a random sub-sample phase. We model all of them by a low-rank, energy-dependent excess covariance. In the whitened frame aligned to the pulse,

$$\boldsymbol{y}_{\mathrm{w}} \sim \mathcal{N}\Big(T_\tau \boldsymbol{\mu}(s),\; I + T_\tau V^{\mathsf{T}} \Sigma(s) V T_\tau^{\mathsf{T}}\Big)\,, \tag{9}$$

with $V \in \mathbb{R}^{K_n \times N}$ orthonormal noise modes and $\Sigma(s)$ a symmetric $K_n \times K_n$ matrix with $\Sigma(0) = 0$ and eigenvalues > -1. In the raw domain, using lemma 1, this is

$$R(E, t_0) = R_0 + T_{t_0}\, G\, \Sigma(E)\, G^{\mathsf{T}}\, T_{t_0}^{\mathsf{T}}, \qquad G = R_0^{1/2} V^{\mathsf{T}}, \tag{10}$$

which is the form of the original proposal. Because T_{t_0} commutes with R_0, the Gram matrix $G^{\mathsf{T}} T_{t_0}^{\mathsf{T}} R_0^{-1} T_{t_0} G = V V^{\mathsf{T}} = I$ does not depend on t_0. This is why the shift does not spoil the low-rank algebra. Negative eigenvalues of Σ are allowed: in a saturating TES the noise during a pulse can be *lower* than the baseline noise (section 7).

5.2 Reduction to K_b dimensions

Theorem 1. *Let $\tilde{B} \in \mathbb{R}^{K_b \times N}$ be an orthonormal basis of* $\mathrm{span}(B, V)$*, $\boldsymbol{x}(\tau) = \tilde{B} T_{-\tau} \boldsymbol{y}_{\mathrm{w}}$, $\boldsymbol{m}(s) = \tilde{B} B^{\mathsf{T}} \boldsymbol{c}(s)$, $Q = \tilde{B} V^{\mathsf{T}}$ and $M(s) = \Sigma(s)\,[I + \Sigma(s)]^{-1}$. Then*

$$-2 \log L(s, \tau) = \|\boldsymbol{y}_{\mathrm{w}}\|^2 - \|\boldsymbol{x}\|^2 + \|\boldsymbol{x} - \boldsymbol{m}\|^2 - (\boldsymbol{x} - \boldsymbol{m})^{\mathsf{T}} Q M Q^{\mathsf{T}} (\boldsymbol{x} - \boldsymbol{m}) + \log\det(I + \Sigma) + \textit{const.} \tag{11}$$

Proof. Let $U = T_\tau V^{\mathsf{T}}$, which has orthonormal columns. The Woodbury identity [22, 23] gives $(I + U\Sigma U^{\mathsf{T}})^{-1} = I - U(\Sigma^{-1} + I)^{-1} U^{\mathsf{T}} = I - UMU^{\mathsf{T}}$; the last form holds also for singular Σ. The determinant lemma gives $\det(I + U\Sigma U^{\mathsf{T}}) = \det(I + \Sigma U^{\mathsf{T}} U) = \det(I + \Sigma)$. For the residual $\boldsymbol{r} = \boldsymbol{y}_{\mathrm{w}} - T_\tau \boldsymbol{\mu}$, orthogonality of T_τ gives $\boldsymbol{r}^{\mathsf{T}} \boldsymbol{r} = \|T_{-\tau} \boldsymbol{y}_{\mathrm{w}} - \boldsymbol{\mu}\|^2$ and $U^{\mathsf{T}} \boldsymbol{r} = V(T_{-\tau} \boldsymbol{y}_{\mathrm{w}} - \boldsymbol{\mu})$. Decompose $T_{-\tau} \boldsymbol{y}_{\mathrm{w}}$ into its projection $\tilde{B}^{\mathsf{T}} \boldsymbol{x}$ and an orthogonal remainder. Since $\boldsymbol{\mu} = \tilde{B}^{\mathsf{T}} \boldsymbol{m}$ and $V = Q^{\mathsf{T}} \tilde{B}$, we get $\|T_{-\tau} \boldsymbol{y}_{\mathrm{w}} - \boldsymbol{\mu}\|^2 = \|\boldsymbol{y}_{\mathrm{w}}\|^2 - \|\boldsymbol{x}\|^2 + \|\boldsymbol{x} - \boldsymbol{m}\|^2$ and $V(T_{-\tau} \boldsymbol{y}_{\mathrm{w}} - \boldsymbol{\mu}) = Q^{\mathsf{T}}(\boldsymbol{x} - \boldsymbol{m})$. □

Cost. Whitening and one real FFT cost $\mathcal{O}(N \log N)$ per record. The aligned projections $\boldsymbol{x}(\tau)$ cost $\mathcal{O}(K_b N \log N)$ for all integer lags at once, or $\mathcal{O}(K_b N)$ at one continuous τ with its first two derivatives (eq. (5)). The rest of (11) costs $\mathcal{O}(K_b K_n + K_n^3)$ per (s, τ). A dense non-stationary filter instead needs an $\mathcal{O}(N^3)$ factorisation of $R(E, t_0)$ for every energy, and for every t_0 if the shift is applied to the covariance. Alternatively it can precompute inverses on a grid at $\mathcal{O}(N^2)$ memory per node. Here $K_b \le K + K_n \approx 12$ for $N = 1024$.

5.3 Learning V and $\Sigma(s)$ and what is identifiable

Given Stage A/B estimates $(\hat{s}_j, \hat{\tau}_j)$, the aligned whitened records $\boldsymbol{z}_j = T_{-\hat{\tau}_j} \boldsymbol{y}_{\mathrm{w}}^{(j)}$ supply residuals. We take V as the top-K_n eigenvectors of the pooled residual covariance. Σ is estimated at knots s_b as $V \widehat{\mathrm{Cov}}_b(\boldsymbol{z}) V^{\mathsf{T}} - I$, with $\Sigma(0) = 0$, and interpolated with a monotone cubic (PCHIP) per entry, with eigenvalues clipped at $-\frac{1}{2}$. The definition of the residual matters.

Proposition 5 (Tangential fluctuations are not identifiable from continuum data)**.** *Let records at energy E fluctuate as $\boldsymbol{\mu}(s) + \xi\, \boldsymbol{d}$ with ξ zero-mean and $\boldsymbol{d} = \alpha\, \boldsymbol{\mu}'(s) + \boldsymbol{d}_\perp$, $\boldsymbol{d}_\perp \perp \boldsymbol{\mu}'(s)$. To first order in ξ, the distribution of records is the same as that of a model with fluctuation $\xi \boldsymbol{d}_\perp$ and an energy distribution $p(s)$ convolved with the law of $\alpha\xi$. If p is unknown and smooth (a*

Algorithm 1 Template-free optimal pulse processing

Require: pulse records $\{\boldsymbol{y}_j\}$, pulse-free records, K, K_n, number of lines

$\lambda_k \leftarrow$ periodogram of pulse-free records; $\boldsymbol{y}_\mathrm{w} \leftarrow W\boldsymbol{y}$ ▷ lemma 1

$B \leftarrow$ top-K SVD of $\{\boldsymbol{y}_\mathrm{w}^{(j)}\}$ with SNR $>$ SNR$_\mathrm{min}$

repeat ▷ Stage A

  $\tau_j \leftarrow \arg\max_\tau \|B_{1:r} T_{-\tau} \boldsymbol{y}_\mathrm{w}^{(j)}\|^2$ (FFT lags, then Newton)

  $B \leftarrow$ top-K right singular vectors of $\{T_{-\tau_j}\boldsymbol{y}_\mathrm{w}^{(j)}\}$; re-gauge

until J_r converged

$\boldsymbol{a}_j \leftarrow BT_{-\tau_j}\boldsymbol{y}_\mathrm{w}^{(j)}$; fit spline graph $\boldsymbol{c}$; $s \leftarrow$ arc length ▷ Stage B

$(s_j, \tau_j) \leftarrow$ stationary ML (eq. (11) with $\Sigma = 0$)

clusters $\leftarrow$ peaks of $\{s_j\}$; $V, \Sigma(s) \leftarrow$ cluster residual covariances ▷ Stage C

$(s_j, \tau_j) \leftarrow \arg\min$ eq. (11); $E_j \leftarrow E(s_j)$ from reference lines

continuum), the two models cannot be distinguished. If p has atoms (mono-energetic lines), the convolution is detectable and α is identified.

Proof. $\boldsymbol{\mu}(s) + \xi\alpha\boldsymbol{\mu}'(s) + \xi\boldsymbol{d}_\perp = \boldsymbol{\mu}(s + \alpha\xi) + \xi\boldsymbol{d}_\perp + \mathcal{O}(\xi^2)$. □

Consequently, residuals about the fitted curve, $\boldsymbol{z}_j - B^\mathsf{T}\boldsymbol{c}(\hat{s}_j)$, reveal only $\boldsymbol{d}_\perp$; we call this label-free variant "curve". Since $\boldsymbol{d}_\perp \perp \boldsymbol{\mu}'$, down-weighting it does not change the energy estimate to first order. The "line-aware" variant instead takes residuals about the mean of each mono-energetic cluster, found as a peak of the histogram of $\hat{s}$. This reveals the full $\boldsymbol{d}$, including the tangential part that broadens the line. The resulting ML estimate is the oblique projection along $\boldsymbol{d}$ (cf. Ref. [7]). It is a learned, multi-dimensional generalisation of the empirical rise-time and arrival-phase corrections used in practice [4]. No energies are needed for this step, only the existence of sharp lines. Lines with a poor peak-to-continuum ratio contaminate the cluster covariance with genuine energy spread.

5.4 Optimisation and Cramér–Rao bounds

We maximise (11) jointly over (s, τ) with a Levenberg-damped 2×2 Newton iteration and backtracking, vectorised over records. Derivatives in τ are analytic, via (5); derivatives in s are central differences of the K_b-dimensional expression. The iteration starts from the Stage A/B estimates. The Fisher information of the learned model is [24]

$$I_{ij} = \partial_i\boldsymbol{\mu}^\mathsf{T} R_w^{-1}\partial_j\boldsymbol{\mu} + \tfrac{1}{2}\,\mathrm{tr}\Big(R_w^{-1}\partial_i R_w\, R_w^{-1}\partial_j R_w\Big), \quad \partial_\tau R_w = -(DV^\mathsf{T}\Sigma V + V^\mathsf{T}\Sigma V D^\mathsf{T}), \tag{12}$$

with $\partial_s R_w = V^\mathsf{T}\Sigma'(s)V$. We report bounds with and without the covariance term, because the latter relies on the exact energy dependence of Σ.

Algorithm 1 summarises the procedure.

6 Physical simulation

To test the method on effects that are physical rather than put in by hand, we integrate the coupled electro-thermal Langevin equations of a voltage-biased one-body TES [2]:

$$C\,\dot{T} = -K(T^n - T_b^n) + I^2 R(T) + P_\mathrm{dep}(t) + \delta P_\mathrm{ph} - I\,\delta V_J, \tag{13}$$

$$L\,\dot{I} = V_b - I\,[R_L + R(T)] + \delta V_J + \delta V_L, \tag{14}$$

with $R(T) = \frac{1}{2}R_n\{1 + \tanh[(T - T_c)/w]\}$. The noise sources are white with one-sided spectral densities $4k_B T R(T)$ (TES Johnson), $4k_B T_b R_L$ (shunt Johnson) and $4k_B T^2 G(T) F$ (thermal link,

Table 1: Simulation parameters and derived bias-point quantities.

T_c, T_b	100 mK, 55 mK	C, $G(T_c)$, n	1.2 pJ/K, 250 pW/K, 3
w, R_n, R_0/R_n	1.4 mK, 10 mΩ, 0.30	R_L, L	0.3 mΩ, 40 nH
F, SQUID noise	0.6, 6 pA/$\sqrt{\text{Hz}}$	$\tau_{\text{th},0}$, κ	12 µs, 0.02 (0 in scenario b)
f_s, N, pre-trigger	200 kHz, 1024, 256	arrival jitter	$\mathcal{U}(-2, 2)$ samples
α, loop gain	99, 27.5	τ_{eff}, I_0	170 µs, 47.6 µA

$G = nKT^{n-1}$). They are evaluated at the *instantaneous* T and R, so the noise is genuinely non-stationary during a pulse. A white SQUID current noise is added at the output. The energy is deposited through an absorber thermalisation profile $P_{\text{dep}} = (E/\tau_{\text{th}})e^{-(t-t_0)/\tau_{\text{th}}}\Theta(t - t_0)$, integrated exactly over each time step so that arrival times are continuous. Position dependence enters through $\tau_{\text{th}} = \tau_{\text{th},0}(1 + \kappa\eta)$ with $\eta \sim \mathcal{U}(-1, 1)$. We integrate with the Euler–Maruyama method [25], using eight sub-steps per output sample and a 1 ms noise burn-in before every record. The ADC is modelled as a boxcar average over each sample period, so the residual aliasing of the pulse onset is present, as in a real digitiser. Table 1 lists the parameters.

Spectra and evaluation protocol. Each scenario has a training spectrum of 2×10^4 unlabelled events: 60% in lines at 1.5, 3.0, 4.5, 5.9, 8.0, 10.0 and 12.0 keV and 40% in a flat continuum from 0.5 to 13 keV. A statistically independent test spectrum of the same size has lines at 1.0, 2.2, 3.7, 5.0, 6.5, 7.2, 9.0 and 11.0 keV, all *between* the training lines. This tests the interpolation of the manifold and of $\Sigma(s)$. There are also 3000 noise records. All models are learned on the training set only. To compare estimators independently of calibration interpolation, we convert each estimator's line width to eV with its own local gain $d\langle\hat{x}\rangle/dE$. The gain is measured by local quadratic regression on the test continuum within ±600 eV of the line. FWHM values are 2.355× the clipped (4σ) standard deviation. Timing jitter is the robust standard deviation of $\hat{t}_0 - t_0$ within each line. A common offset (time walk) is removed per line; that offset is a gauge choice (section 8). The two scenarios are (a) $\kappa = 0.02$ (a ±2% thermalisation-time spread) and (b) $\kappa = 0$.

Estimators compared. (1) OF with arrival-time search and a template averaged from the 5.9 keV training line. (2) The norm of the Stage A coefficient vector, in the spirit of Ref. [12]. (3) Stage B closest-point projection. (4) Joint (s, τ) ML with stationary noise. (5) Woodbury ML with label-free ("curve") covariance. (6) Woodbury ML with line-aware covariance. As references we use the CRB of the learned models and an oracle: a local generalised least squares estimator whose covariance (K_n modes) is estimated on half of the *test* line itself and applied to the other half (cross-fitted). Unless stated otherwise, $K = 4$, $r = 1$, $K_n = 8$, $\text{SNR}_{\text{min}} = 30$ and a search window of ±8 samples.

7 Results

7.1 Pulses and noise

Figure 1 shows the simulated detector. The peak current per unit energy falls by a factor of 2.0 between 1 and 12 keV, while the decay lengthens. This is the shape evolution that biases and degrades a fixed-template OF. Panel (d) shows that the noise variance during the pulse departs strongly from baseline. It falls *below* baseline while the sensor is driven up the transition: the loop gain and the current responsivity drop, so the same noise sources produce less current noise. This is mainly a change of the electro-thermal *response*, not of the sources. With the source amplitudes frozen at their bias-point values (dotted) the drop is even larger, because the hotter, more resistive sensor then no longer partly compensates with larger Johnson and phonon noise.

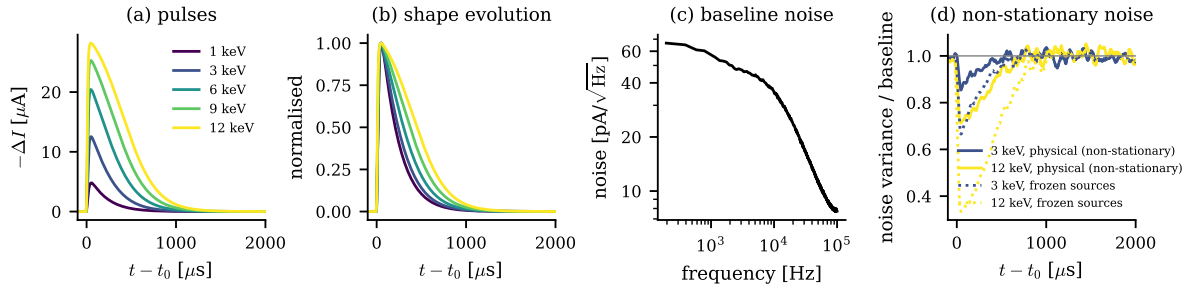


Figure 1: Simulated TES. (a) Mean pulses at 1–12 keV. (b) The same, normalised, showing saturation and the lengthening decay. (c) Baseline noise spectrum. (d) Noise variance during the pulse relative to baseline (1500 records per curve), for the physical model (solid) and with the noise-source amplitudes frozen at the bias point (dotted).

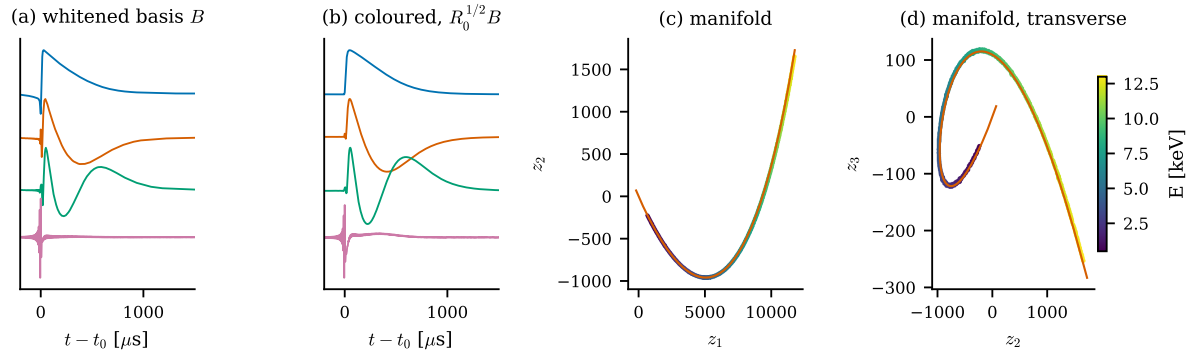


Figure 2: (a) Whitened shift-invariant basis ($K = 4$). (b) The same vectors in the raw domain. (c, d) Aligned coefficients of the training spectrum in the rotated frame of the manifold, coloured by true energy, with the fitted curve.

7.2 Learned subspace and manifold

Stage A converged in 3 iterations (fig. 2). The leading whitened basis vector is a sharpened mean pulse. The next ones encode the lengthening decay and the changing rise, and the fourth vector holds the high-frequency structure of the onset. Coloured back with $R_0^{1/2}$, they become the mean pulse and its shape-variation modes in current units. The coefficients trace a smooth curve whose transverse excursions grow with energy. The spline graph fits it with a GCV-selected penalty, and its arc length reaches $s_{\max} = 12389$ (the SNR at about 13 keV).

7.3 Structure of the pulse-correlated noise

Figure 3(a) shows the eigenvalue spectrum of the residual covariance about the cluster means. A few eigenvalues stand far above the Marchenko–Pastur edge [26] of the noise bulk, so the excess covariance is genuinely low-rank. Correlating the mode amplitudes with the simulation truth identifies the leading modes. The two largest (pooled eigenvalues ≈ 295 and 256) are a pair driven by the sampling phase: their amplitudes follow $\cos 2\pi t_0$ and $\sin 2\pi t_0$ with $|r| \approx 0.95$. This is the residual aliasing of the pulse onset, which no Fourier-shifted template can represent but which the low-rank model captures automatically. The next two (≈ 82 and 4) are driven by the thermalisation parameter η ($|r| = 0.8$–0.99). There are two of them because the direction of the rise-time effect rotates with energy. The physical non-stationary noise of fig. 1(d) shows up mainly as directions of *reduced* variance (pooled eigenvalues ≈ 0.4), which a variance-ranked model does not select (see the ablation below).

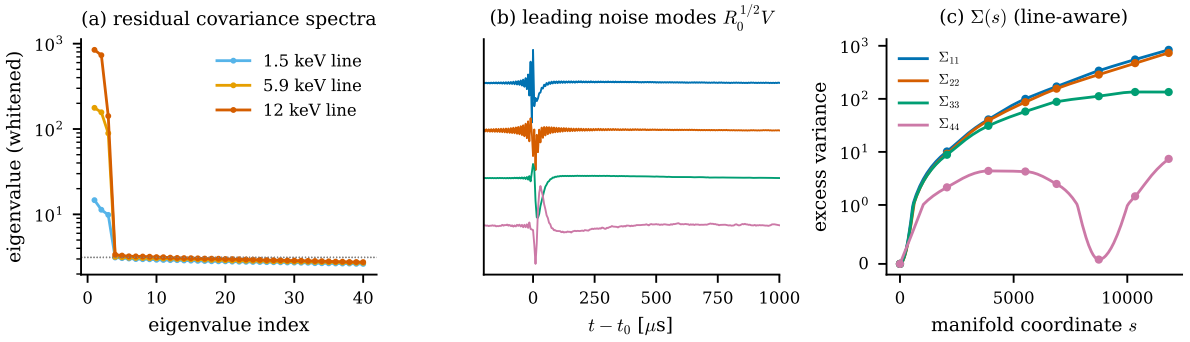


Figure 3: (a) Eigenvalues of the whitened residual covariance of three training lines; dotted: Marchenko–Pastur upper edge for pure noise. (b) Leading noise modes in the raw domain. (c) Diagonal of the interpolated excess covariance $\Sigma(s)$ (symlog scale), with the line knots.

7.4 Energy resolution

Figure 4 and table 2 summarise the energy resolution on the held-out test lines. In scenario (a), the OF degrades from 2.0 to 12.4 eV between 1 and 11 keV, and the PCA norm behaves similarly. The stationary manifold ML removes the template-mismatch penalty, reaching 6.0 eV at 11 keV. Its resolution is then limited by the tangential part of the rise-time and sampling-phase fluctuations. As proposition 5 predicts, the label-free Woodbury variant leaves this limit essentially unchanged. The line-aware Woodbury ML removes it and reaches 1.7–2.8 eV, an improvement over the OF of 1.2× at 1 keV, 3.3× at 6.5 keV and 4.5× at 11 keV. It agrees with the oracle local Gaussian model to within 0.03 eV at every test line. That oracle uses the test line's own covariance; ours is interpolated from training lines at other energies. Scenario (b), where only non-linearity and non-stationary noise remain, is sobering. The OF is already good (3.2 eV at 11 keV against 2.8 eV for the line-aware ML), and the gain is at most 15%. A one-parameter family of shapes is handled well by a single template plus a non-linear gain curve. The efficiency loss is only the cosine between template and local tangent. The stationary manifold ML is even slightly *worse* than the OF at the highest energies (3.6 eV). Its projection direction, the local tangent, is more exposed to the tangential part of the sampling-phase modes than the OF template is. The line-aware covariance removes this exposure. In short, the large gains in scenario (a) come from modelling shape fluctuations, which a fixed template cannot do. They do not come from the non-linearity as such.

The learned-model CRBs lie slightly *above* the achieved widths, so they are not bounds on the physical process. The stationary CRB assumes baseline noise along the manifold tangent. In the simulation the noise during the pulse is lower than baseline, for the electro-thermal reason of fig. 1(d). At 1 keV the achieved width implies a tangential noise variance of about 0.87 of baseline. The variance-ranked low-rank model adds excess variance (the rise-time and sampling-phase modes) but does not represent these directions of reduced variance, so its CRB is, if anything, larger than the stationary one. Ranking modes by the likelihood gain captures some of the reduction when it is pooled over many events (see the ablation). On the ∼700 events of half a line, however, it overfits the small sample eigenvalues of the Marchenko–Pastur bulk and degrades the oracle badly. We therefore use the oracle (same model class, $K_n = 8$ variance-ranked modes, estimated on the test line itself) as the practical reference. Our estimator, which interpolates $\Sigma(s)$ from training lines at other energies, reaches it.

7.5 Timing

Figure 5 compares the arrival-time jitter with the CRB. Without morphology variation (b), the jitter equals the CRB at 1 keV (30 ns against 29 ns). At higher energies it levels off at 18–30 ns,

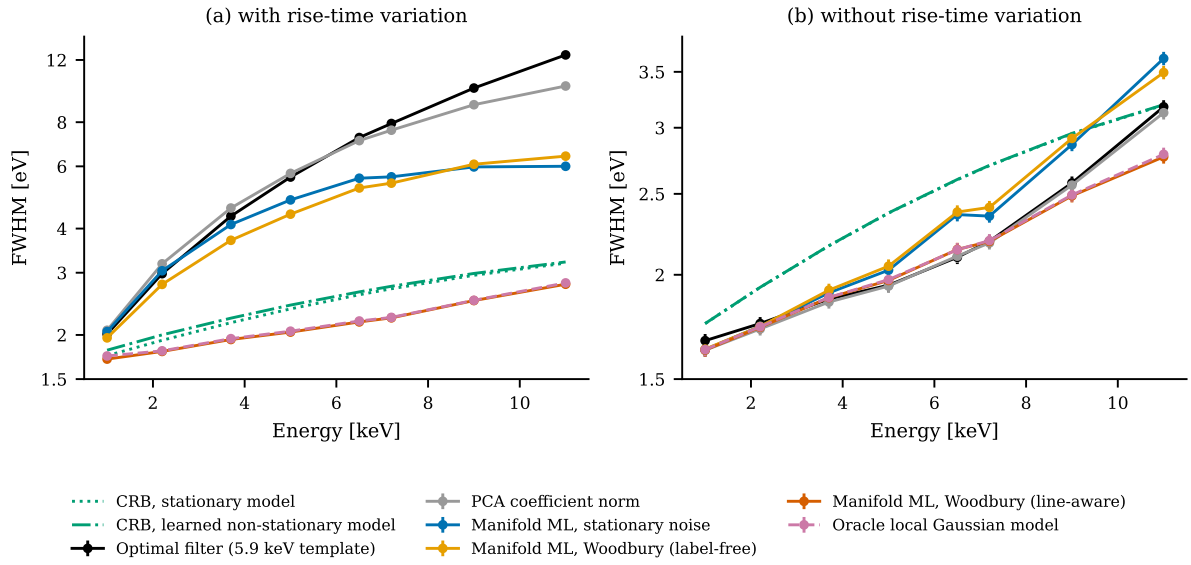


Figure 4: Energy resolution (FWHM) on held-out test lines that lie between the training lines. (a) With ±2% thermalisation-time spread; (b) without. Error bars: statistical. Dotted and dash-dotted: CRB of the learned stationary and non-stationary models. These are not bounds on the simulated process, whose in-pulse noise is partly below baseline (see text).

while the bound keeps falling to 3.3 ns. The floor is deterministic. Regressing the timing error on three harmonics of the *true* sub-sample phase removes it, and the remaining jitter is within a factor 1.11 of the CRB at every line (3.6 ns at 11 keV). The cause is the same residual aliasing of the pulse onset that produces the sampling-phase noise modes: a pulse that is not band-limited is displaced differently depending on where it falls between samples. The low-rank covariance treats these modes as random, which is correct for the energy estimate, but the phase-dependent displacement is a *bias* in t_0. We tried to correct it without labels by equalising the histogram of $\hat{t}_0$ mod 1 within bins of $\hat{s}$ (raw and Fourier-smoothed). This failed in our tests: within a bin the energy-dependent walk smears the phase histogram, and the correction added more scatter than it removed. A phase-resolved manifold $\boldsymbol{c}(s, \varphi)$ is the natural remedy and is left for future work. With morphology variation (a), all estimators sit at 56–89 ns: a ±2% change of the thermalisation time moves the apparent arrival time by more than the noise does. Under shape variation the definition of t_0 is itself a gauge choice (section 8).

7.6 Low-energy behaviour

Figure 6 shows the response to events from 0 to 320 eV. Each estimator is calibrated by a quadratic through the origin fitted at 80–320 eV, because the detector is slightly non-linear even there. The coefficient norm is biased upward by 1.4 eV at zero energy, and the bias decays only over the first few tens of eV, as proposition 4 predicts. The OF with an arrival-time search is also biased (0.36 eV): maximising the filter output over lags selects positive noise excursions. The manifold ML is unbiased (0.05 eV), because its search over τ is symmetric in the sign of s. The price is a 34% wider, bimodal zero-energy distribution, since the free arrival time picks the largest $|s|$ of either sign. Above a few eV all three have the same spread.

7.7 Computational cost

Figure 7 compares the time per likelihood evaluation of the K_b-dimensional formula (11) with a dense Cholesky evaluation of the same non-stationary model. The two agree to a relative

Table 2: FWHM energy resolution [eV] on the held-out test lines.

Estimator E [keV]	1.0	2.2	3.7	5.0	6.5	7.2	9.0	11.0
Scenario with rise-time variation								
Optimal filter (5.9 keV template)	2.01	2.98	4.34	5.60	7.24	7.93	9.99	12.40
PCA coefficient norm	2.06	3.18	4.57	5.73	7.10	7.59	8.96	10.13
Manifold projection (Stage B)	2.04	3.04	4.11	4.82	5.55	5.60	5.98	6.00
Manifold ML, stationary noise	2.04	3.04	4.11	4.82	5.55	5.60	5.98	6.00
Manifold ML, Woodbury (label-free)	1.96	2.78	3.71	4.39	5.21	5.38	6.08	6.41
Manifold ML, Woodbury (line-aware)	1.71	1.80	1.94	2.04	2.18	2.24	2.50	2.78
Oracle local Gaussian model	1.74	1.80	1.95	2.05	2.19	2.24	2.51	2.81
CRB, learned model	1.81	2.00	2.23	2.43	2.65	2.75	2.99	3.22
Scenario without rise-time variation								
Optimal filter (5.9 keV template)	1.67	1.75	1.87	1.94	2.10	2.19	2.58	3.18
PCA coefficient norm	1.62	1.72	1.86	1.94	2.11	2.18	2.56	3.13
Manifold projection (Stage B)	1.63	1.74	1.90	2.03	2.36	2.35	2.86	3.63
Manifold ML, stationary noise	1.63	1.74	1.90	2.03	2.36	2.35	2.86	3.63
Manifold ML, Woodbury (label-free)	1.63	1.73	1.92	2.05	2.38	2.41	2.91	3.49
Manifold ML, Woodbury (line-aware)	1.63	1.73	1.88	1.97	2.14	2.19	2.49	2.77
Oracle local Gaussian model	1.63	1.73	1.88	1.97	2.14	2.20	2.49	2.79
CRB, learned model	1.75	1.93	2.17	2.37	2.60	2.70	2.95	3.20

precision of 10^{-16} for all N. At $N = 1024$ the Woodbury evaluation is 2204 times faster per (E, t_0), and the advantage grows with N. The full pipeline processes 688 records/s on one CPU core for the line-aware ML, against 2042 for the OF, in our pure-NumPy implementation.

7.8 Ablations

Figure 8 varies K and K_n. The resolution is insensitive to K between 2 and 8: the manifold is smooth enough that two whitened components describe it, and extra components neither help nor hurt. With $K_n \leq 2$ the model holds only the sampling-phase pair and the resolution stays at the stationary level. From $K_n = 4$, the rise-time mode is included, the resolution drops to the oracle value, and adding more modes neither helps nor hurts noticeably. We also tried ranking modes by the Gaussian likelihood gain $\lambda - 1 - \log\lambda$ instead of by variance. This keeps directions of *reduced* noise ($\lambda \approx 0.4$) and improves the highest-energy lines by 1–4%, at a cost of about 2% at low energy. We kept the simpler variance ranking as the default.

8 Discussion

What the simulations support. With a ±2% thermalisation spread, the template-free, line-aware estimator improves the FWHM over a fixed-template OF by 1.2× to 4.5× on held-out lines of a saturating TES. The largest gains appear where saturation is strongest. It matches a per-line oracle Gaussian model to within 0.03 eV. The low-energy biases of the coefficient norm and of the OF with time search are removed. Without morphology variation, timing reaches the CRB once a deterministic sampling-phase bias is removed. The non-stationary likelihood costs about as much as a stationary filter with K_b templates.

What needs qualification. Several statements of the original research plan have to be weakened. First, "calibration independence" holds for the pulse manifold and for the noise model, which are learned from unlabelled data. The absolute energy scale $E(s)$ still needs reference lines, and the tangential part of the covariance needs mono-energetic clusters (proposition 5).

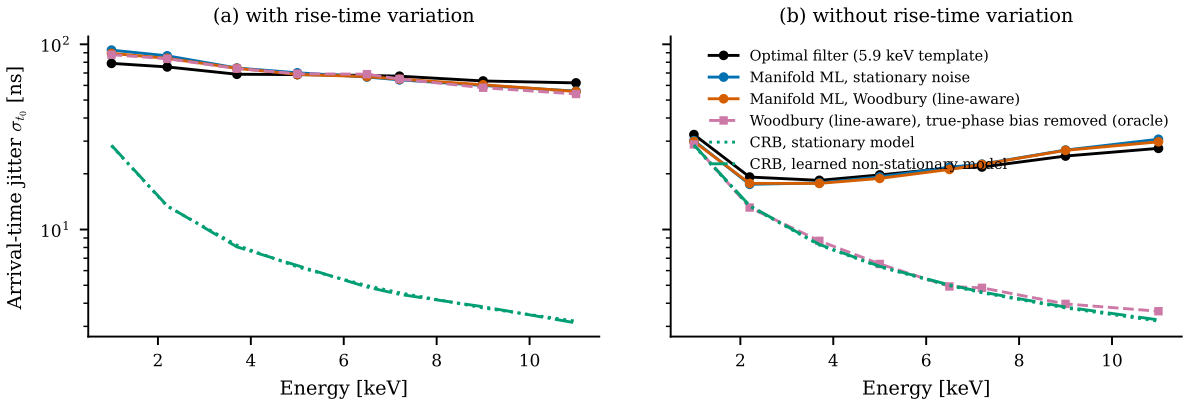


Figure 5: Arrival-time jitter per test line (per-line time-walk offset removed; 1 sample = 5 µs). Dashed: line-aware ML after regressing out the dependence on the true sub-sample phase (oracle diagnostic). Dotted and dash-dotted: CRB of the learned stationary and non-stationary models.

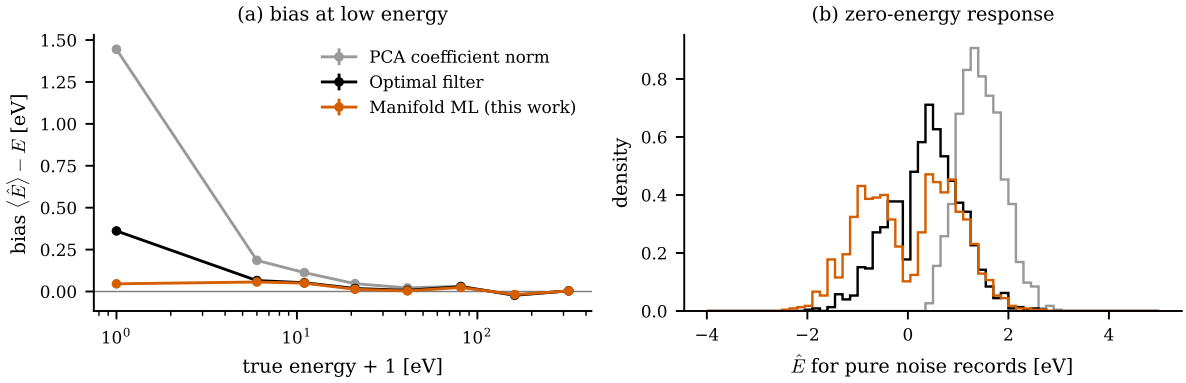


Figure 6: (a) Bias of three estimators at low energy. (b) Distribution of the estimates for pure-noise records.

Second, Cholesky pre-whitening must be replaced by the symmetric spectral root for shift invariance to be exact (lemma 1). Third, the cost of the non-stationary likelihood is $\mathcal{O}(N \log N)$ per record for the transform, plus $\mathcal{O}(K_b N)$ per continuous (E, t_0) evaluation (or $\mathcal{O}(K_b K_n)$ once the projections are tabulated). It is not $\mathcal{O}(N \log N)$ for the inverse itself, which is never formed. Fourth, the CRB of the learned low-rank model is not a bound for the physical process. The physical noise is *reduced* during the pulse in directions the variance-ranked model does not represent, so the estimator beats that "bound". A tighter reference needs a better-conditioned estimate of the full in-pulse covariance than a few hundred events per energy allow. Fifth, the "$> 2\times$ FWHM improvement in saturated regimes" holds only when shape fluctuations such as position dependence or sampling-phase effects are present. For a detector whose pulses form a strict one-parameter family, a fixed template with a non-linear gain curve loses at most 15% in our simulation. Sixth, sub-sample timing "locked to the Fisher limit" needs a correction of the sampling-phase bias, which we achieved only as an oracle.

Timing gauge. With energy-dependent shapes, "arrival time" is defined only up to an energy-dependent offset $\delta(E)$ (time walk): the reparameterisation $\boldsymbol{\mu}(s) \mapsto T_{\delta(s)}\boldsymbol{\mu}(s)$, $t_0 \mapsto t_0 - \delta(s)$ leaves the likelihood unchanged. Our gauge aligns the mean-pulse direction. A Fisher-orthogonal gauge, in which $\langle \boldsymbol{\mu}', D\boldsymbol{\mu} \rangle = 0$, would decorrelate energy and time estimates. A physical onset gauge needs an external timing reference. Because the noise-free manifold is known, conversion

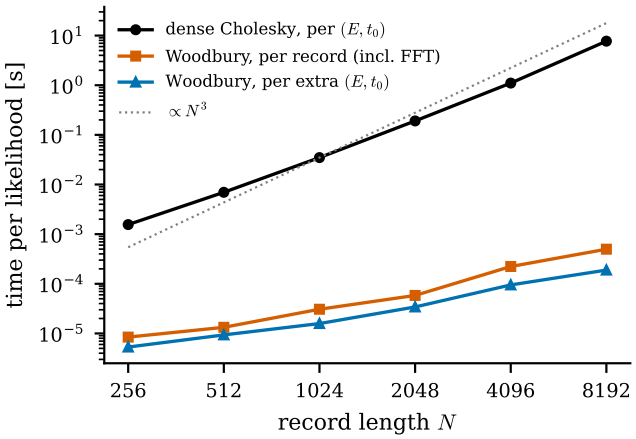


Figure 7: Time per evaluation of the non-stationary log-likelihood: dense Cholesky versus theorem 1 ($K_s = 4$, $K_n = 8$).

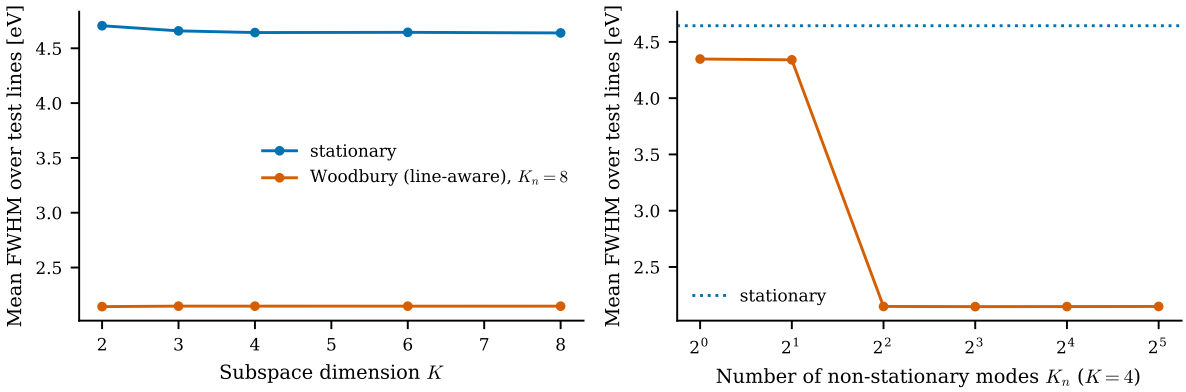


Figure 8: Mean FWHM over the test lines versus the subspace dimension K (left) and the number of non-stationary modes K_n (right), scenario (a).

between gauges is a deterministic post-processing step.

Limitations and outlook. Our results are from simulation. The real-data tests that matter most are TES arrays with pile-up, gain drift and non-Gaussian tails in the shape fluctuations. Pile-up rejection can use the likelihood value (11) directly. Gain drift enters as a slow reparameterisation of s and could be tracked by line centroids. Continuous morphology parameters could be estimated explicitly with a two-dimensional manifold $\boldsymbol{c}(s, \eta)$; we did not pursue this because the tangential identifiability problem applies there too, and the line-aware covariance already reaches the oracle. The low-rank model would fail for noise whose non-stationarity is spread over many modes. A diagonal-plus-low-rank model in the time domain would keep part of the speed-up.

9 Conclusion

We have presented a template-free pulse processing chain for non-linear microcalorimeters. Its three stages are shift-invariant PCA in the whitened domain, an arc-length spline manifold, and a Woodbury-accelerated non-stationary maximum likelihood. Each stage is exact in the sense that it optimises a stated likelihood. The chain reaches the resolution of an oracle Gaussian

model on held-out lines of a physically simulated, saturating TES, at the cost of a small bank of templates. The open-source implementation is intended to make these methods easy to test on real detector data.

Data and code availability

All code, including the simulator and the scripts that produce every figure and number in this paper, is available at https://github.com/ytcs/empca.

A Software

The package `empca` (import name) implements every component as a separate, tested class: `StationaryNoise`, `ShiftInvariantPCA`, `PulseManifold`, `LowRankNoise`, `ManifoldFitter` and `OptimalFilter`. A pipeline object chains them:

```
from empca import PulseProcessor
proc = PulseProcessor(n_components=4, n_lines=7, n_noise_modes=8)
proc.fit(train_records, noise_records)        # unlabelled
res = proc.transform(records)                  # res.s, res.t0, res.nll
proc.calibrate(proc.line_centers_, line_energies_eV)
energy = proc.energy(res.s)
fwhm = proc.resolution(res.s)                  # learned-model CRB
```

The frequency-domain EM-PCA of Ref. [12] remains available as `empca.EMPCA`.

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